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Comparison and Correction of the Light Sensor Output from 48 Wearable Light Exposure Devices by Using a Side-by-Side Field Calibration Method

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ABSTRACT Measurement of personal light exposures and activity has gained popularity in studies of the circadian rhythm and its effects on human health. Calibration of a batch of measuring devices may be needed, especially before initiating interventional studies, but manufactory calibration of devices before every initiated study is costly for the researcher and therefore often left out. Still, knowledge of inter-equipment variability is essential and seldom provided by the manufactory. The aim of the present study was to develop and test a method for field calibration of Actiwatch Spectrum devices. We tested 48 Actiwatch devices side by side under various light sources and present the red, green, blue, and white light response variability among the Actiwatches. The influence of different spatial and spectral light environments on the white light response when compared with the output from a calibrated photometer is discussed. In agreement with previous studies by Price and others [2012] and Figueiro and others [2013], we confirm the devices' white light responses to be highly dependent on both the spatial and the spectral composition of the light. The white light response represents photopic illuminance only to a minor degree and light source-specific calibration may therefore be needed in some cases. Moreover, light responses were found to vary between devices by up to 60%. Implications are that the results of light effects on health issues in studies using Actiwatches are blurred by the equipment variability. To compensate for inter-equipment variability we stress the need for a field calibration procedure. When light exposure devices of lower grade quality are used in spectrally and spatially changing light environments, daylight from a diffused overcast sky is suggested to be used for side-by-side calibration of Actiwatches and similar personal light exposure devices. We suggest that the calibration methods presented can be used for calibration of other practical field devices, with respect to the various sensors already on the market and devices that will be introduced in the future.

KEYWORDS spectral power distribution (SPD), RGB light, scotopic vision, photopic vision, photopic illuminance, illuminance meter

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1. INTRODUCTION

Many different measuring devices for personal light exposure are used in the field. These devices often report photopic illuminance in lux, but the devices seldom meet the requirements of laboratory photometers [Figueiro and others 2013]. Often, the sensors of light exposure devices are not cosine corrected and may not perfectly follow the photopic luminous efficiency function, $V(\lambda)$, as previous work on Philips Respironics Actiwatch Spectrum™ (Philips Healthcare, Best, Netherlands) devices has demonstrated [Figueiro and others 2013; Price and others 2012]. The work of Price and others [2012] made it clear that the specific location, orientation, and shape of light sensors in a measuring device determine the sensors' field of view, thus influencing the response. Although the device dimensions and sensor location may only slightly differ between individual devices, small deviations may have a great impact on the light registration dependent on the present spectral and spatial light conditions. Because of variation between light exposure devices in terms of spectral and spatial light responses, the light intensities being measured depend on the particular light measuring device and the actual light conditions, which are seldom stable. When used in the field, the ever-changing spatial and spectral light condition has consequences for the light measurements of personal light exposure devices. The impact of this is important for comparative studies and stresses the need for high standards of the light sensors used or in-depth knowledge of the lower quality devices used.

When measuring light at a fixed location where the major contribution of light comes from one stable light source, the use of low-quality broad-spectrum light sensors will provide useful measurements of, for example, photopic illuminance if the sensor has been calibrated against a light source like the one at the location. This is called "source-specific calibration." In contrast, when used in continuously changing spatial and spectral light conditions, how do we deal with the broad-spectrum light sensor measurements?

Knowledge of humans' daily exposure to light is still limited [Veitch and Galasiu 2012]. Numerous light sensors of varying quality are used on the market, and devices especially used and designed to measure personal light exposure have various designs. Some are to be fixed on the wrist, whereas others are fixed on the forehead [Figueiro and others 2013; Hubalek and others 2010; Rea 2006]. Exposure devices, like, for example, Actiwatch Spectrum devices, are also able to measure activity [Price and others 2012]. Devices worn on the wrist are more discreet to wear than those fixed on the forehead. The devices fixed on the forehead measure the light coming from the same direction as the eyes, thus being more relevant in specific cases, where discreetness is not required [Rea 2006].

There is a great difference in the spectrum of white light sources like, for example, fluorescent and tungsten halogen light (Fig. 1). Inside buildings there is often a mix of light from different sources, including daylight. Colored light or RGB light is now becoming increasingly common. It offers the advantages of obtaining better color rendering [Jost-Boissard and others 2009] or the possibility of mixing light of different colors or diminishing the intensity of certain wavelength bands with the aim of, for example, enhancing nurses' ability to see veins during blood sampling. Moreover, colors have been shown to be connected with emotions, and the same can be assumed to apply to the color of light [Rautkyla and others 2011]. Therefore, and in addition to measuring illuminance, some knowledge of the spectral composition of light exposure is important in studies of colour impacts.

The spectral sensitivity of the R, G, and B sensor; directional response; linearity of the sensor response; and the use of diffusers to enhance the directional response of Actiwatch Spectrum devices was published earlier [Figueiro and others 2013; Price and others 2012]. Price and others [2012] found that the directional response is far from close to a perfect cosine response, with cosine response errors of approximately 30%–50% in the horizontal plane and approximately 60%–65% in the vertical plane.

Ideally for comparative studies, all sensors should have the same defined standard response pattern as if they were cosine corrected and had a spectral sensitivity matching the response of interests like $V(\lambda)$. However, new sensors with different specifications are continuously introduced on the market, and therefore an easy-to-perform standard calibration procedure in the field is needed to limit the bias of equipment variability.

The aim of the present study was to develop and test a method for calibration of Actiwatch devices in field studies, with respect to the various sensors already on the market and devices that will be introduced in future. The Actiwatch white light output (W_{Acti} light) is reported as being photopic illuminance measured in lux. However, the W_{Acti} light is derived from a not published function of the light received by the Actiwatch sensor measuring light in three wavelength bands named after their associated color perceptions like red (R light), green (G light), and blue (B light). The Actiwatch sensor's relative response per wavelength has been reported for the W_{Acti}, R, G,

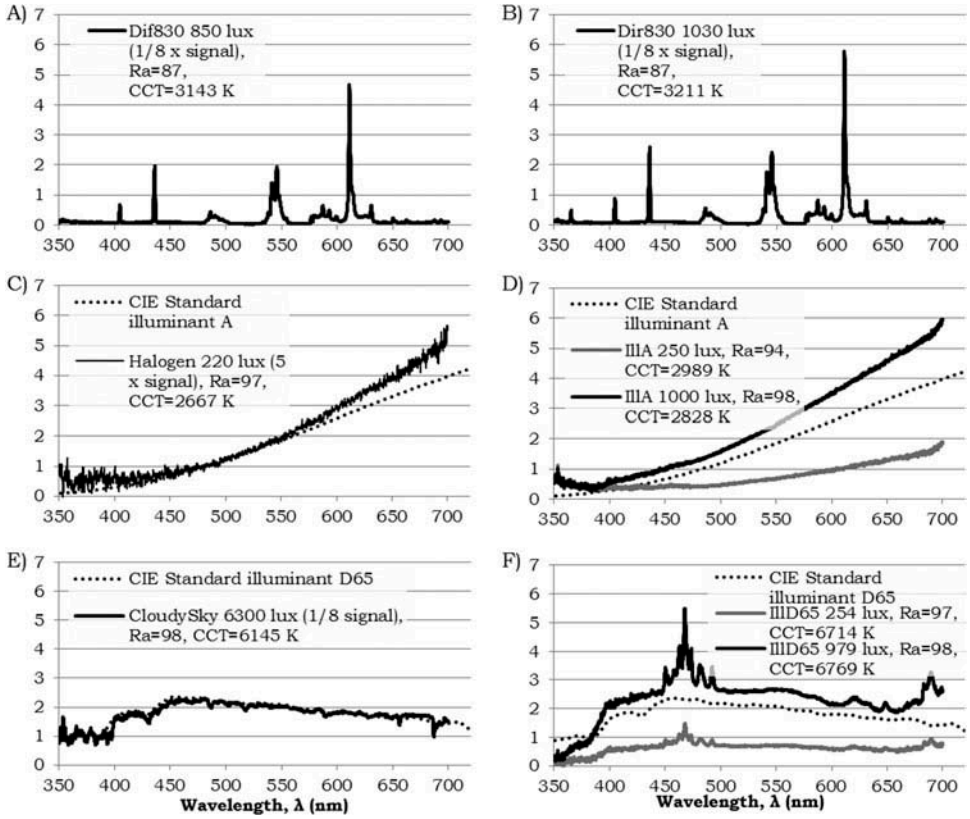


Fig. 1 **Relative spectral composition of the light used for the calibration tests, including (A) diffused fluorescent light (Dif830); (B) direct fluorescent light (Dir830); (C) tungsten halogen (Halogen); and (D) approximate standard illuminant A (IllA), at two light intensities; (E) overcast sky conditions (CloudySky); and (F) approximate standard illuminant D65, at two light intensities. Included in (C) and (D) is the CIE-defined standard illuminant A and in (E) and (F) the CIE defined standard illuminant D65. When indicated in parentheses, the signal responses are corrected linearly to fit the relative scale (*y* axis).**

and B light [Figueiro and others 2013; Price and others 2012]. A defined spectral sensitivity of sensors suggests that when measuring under the same spectral and spatial light conditions, the relationships between the R, G, and B light versus W_{Acti} light should ideally be the same regardless of the particular Actiwatch device used. Moreover, it was presumed that the sensor response was linear with increased intensity. These were the premises when trying to find an easily applicable way to calibrate our Actiwatches responses.

2. METHODS

In a large field study initiated in 2011, the overall objective was to investigate whether the light environment for indoor work and night work has health consequences; for example, to evaluate whether high light exposure at night during night-shift work increases the risk of getting breast cancer. For this purpose, we bought 50 Actiwatch Spectrum devices. The choice was made based on a need for the device to be affordable in order to purchase the numbers needed for the field study, commercially available, discreet to wear, and able to measure both activity and photopic illuminance. In addition, we found it attractive that the Actiwatches were able to measure the R, G, and B spectral characteristics of the light. We examined whether there was predictability and similarity in the RGB and W_{Acti} light responses of 48 Actiwatches (two devices were lost during a field experiment), when measuring the light from typical white light sources.

2.1. Actiwatch Configuring and Management

To configure the Actiwatches for measurements, we used the Actiwatch Communication and Sleep Analysis Software: Respironics Actiware version 5.71.0 (Philips Healthcare, Best, the Netherlands). For the various calibration tests, all Actiwatches were set to log measurements of W_{Acti} light; R, G, and B light; and activity every 15 seconds (epoch = 15 seconds) starting from a fixed point in time. The battery level of Actiwatches was given by the software and varied from "full" to "battery replacement required." Therefore, a test was performed ensuring that

the battery status did not affect the results. The software was also used to retrieve data from the Actiwatches and to create .csv-files for further processing. The Actiwatches are named by serial number.

2.2. Correction Values and Lighting Test Schemes

We wanted to determine the extent of the deviation in the individual devices' light measurement in comparison with that of the other measuring devices; that is, to find one correction value (C_{VAL} = measured value/expected value) of the particular devices' measurement. The correction values used and presented are the least-squares regression inclination value; that is, the reciprocal of the often used multiplicative correction factor. A device with a correction value (C_{VAL}) above one thus measures values too high in relation to the reference sensor photopic illuminance and vice versa, and the percentage deviation can be directly transcribed.

A setup with one light scenario for testing all devices at the same time is referred to in the following as a calibration test. The difference in the average C_{VALs} between different selected light environments using standard light sources was examined.

2.2.1. Light response offset test in complete darkness

An offset test in complete darkness was performed to find the W_{Acti}, R, G, and B light average offset response of the Actiwatches. We used an 8-hour period to calculate the average response, considered to be close to an average sleeping period during one day. The offset test was repeated four times.

2.2.2. Batch of devices being calibrated simultaneously under stable light sources

Several calibration tests were completed under various stable light sources including tungsten halogen light (Test: Halogen, Halogen PAR30 TECH, 75W, General Electric, Fairfield, CT, USA) and fluorescent tube light (4 × Philips Master TL5 HE, 14W/830, Philips Lighting, Pila, Poland). Tests were made in a large and blacked-out room under otherwise stable physical conditions. Two different calibration tests with fluorescent light were performed: Fluorescent light directly received from the light tubes and light being redirected in a diffuse manner, referred to as Dir830 and Dif830, respectively. In Dif830 the light was diffused by a 3-mm frosted polycarbonate plate (PLEXIGLAS, the original from Röhm, Germany), placed 1 cm above the Actiwatches underneath the light source. The distance from the halogen and fluorescent light sources to the Actiwatches was 1.0 m. This distance resulted in a visually even light distribution in the field of the placement of the Actiwatches. Placed next to the Actiwatches were the cosine-corrected detector of a newly purchased reference photometer (Hagner EC1-X, Instr. Nr. 54211, B. Hagner AB, Solna, Sweden) and a cosine corrected detector (CC-3-UV-S, Ocean Optics, Inc., Dunedin, FL, USA) connected to an Ocean Optics HR4000 spectrometer (Ocean Optics, Inc.). The Hagner EC1-X reference lux sensor used was checked in our own lab and matched the response of another newly calibrated reference lux sensor (LMT Photometer B510). The spectrometer was controlled by the software SpectraSuite (SpectraSuite, Spectroscopy Software, Ocean Optics, Inc.). The spectrometer measured the spectrum under which the calibration tests were performed.

2.2.3. Batch of devices being calibrated simultaneously under open sky conditions

Tests under open sky conditions were performed under both a sunny and an overcast sky. The Actiwatches were placed perpendicular to the direction to the sun at noon on a building rooftop in Copenhagen (coordinates: 55.650619, 12.542481), on November 1 and 5, 2013. The test under overcast sky conditions is referred to as CloudySky.

2.2.4. Repeated measurements

The Actiwatches were randomly placed side by side and fixed using a Velcro mounting solution for easy and random rearranging of the Actiwatches below all light sources. At every random arrangement, the Actiwatches were placed side by side in four rows and with the same orientation.

Average C_{VALs} were based on three to eight calibration tests (repetitions) using one particular type of light scenario (Figs. 1A–1C and 1E) and with a new random arrangement of the Actiwatches for each test. To make sure that the time of data logging of the Actiwatches matched the manually transcribed lux readout, a covering black cloth was used, and uncovering and covering the devices indicated the start and end of measurements. For every Actiwatch and calibration test, one C_{VAL} was calculated using an average value based on eight Actiwatch

readout points in time—that is, data during 2 minutes—divided by the mean of 8 corresponding-in-time lux values measured by the reference illuminance sensor. This was done in order to level out sensor noise (electrical noise).

The measured R, G, and B light response per Actiwatch was divided by the mean R, G, and B light response, respectively, for the particular calibration test.

2.2.5. Tests at certified calibration test facility

In addition to testing the Actiwatches side by side simultaneously, calibration tests were performed at a certified calibration test facility at Delta (Venlighedsvej 4, DK-2970 Hoersholm, Denmark). Every Actiwatch was alternately placed for 90 seconds both in a collimated illuminant A approximate light source (IllA) and a D65 approximate daylight source (IlllD65) at two light intensities of approximately 250 and 1000 lux.

The IlllD65 light source was an Osram XBO Xenon Short Arc lamp of 150 W, with a combination of one Schott type LB80, 1.0-mm-thick prefilter and a UV 34, 2.5-mm-thick prefilter. To obtain 250 lux, we further mounted a T0.25 neutral density (ND) filter to reduce the light intensity. This was fitted in an Oriel lamp housing with an adjustable collimating lens used to fine-tune the intensity in the point where the measurement took place. The correlated color temperature was measured to be 6714 K at 250 lux and 6769 K at 1000 lux, respectively.

The IllA light source was a Philips 12V H4 halogen fitted in a lamp housing with a 4.0-mm opal diffuser. The IllA intensity was controlled by metal leaf shutters between the light source and the diffuser. For IllA, the correlated color temperature was measured to be 2989 K at 250 lux and 2828 K at 1000 lux, respectively.

Both lighting systems were fed from a stable DC supply. The spectra under which the calibration tests where made are shown in Figs. 1D and 1F.

We measured illuminance with an LMT photometer with associated detector (light-sensitive surface diameter ø = 30 mm) calibrated and traceable to the Physikalisch-Technische Bundesanstalt (Braunschweig, Germany) and National Physical Laboratory (Teddington, UK). Illuminance was measured at the same position as for the placement of the Actiwatches 1.0 m from the light source.

Correlation between the average W_{Acti} light and the average R, G, and B light for IlllD65 and IllA was made in the same way as for Dif830, Dir830, Halogen, and CloudySky.

Average W_{Acti} and R, G, and B light C_{VALs} for the two light intensities for IlllD65 and IllA, respectively, were calculated per Actiwatch. These C_{VALs} were used in the comparisons of calibration methods.

2.2.6. Comparing different calibration methods

The average W_{Acti} and R, G, and B light C_{VAL} for the two light intensities for IlllD65 and IllA, respectively, was considered to be the result of a standard calibration procedure or method for calibration of devices. This result was compared with the result of the alternative method for calibration, measuring the light simultaneously with all of the devices (repeated measurements of an Actiwatch batch of devices). We compared the C_{VAL} test result of the light scenario that corresponded the most with the spectrum of IllA and IlllD65, thus comparing Halogen versus IllA and CloudySky versus IlllD65.

The following plots were made: (A) average W_{Acti} light C_{VALs} for Halogen and IllA; (B) average R, G, and B light C_{VALs} for Halogen and IllA; (C) average W_{Acti} light C_{VALs} for CloudySky and IlllD65; and (D) average R, G, and B light C_{VALs} for CloudySky and IlllD65. The slope and intercept of the associations and their confidence intervals were estimated using a linear functional model and Deming regression [Deal and others 2009; Mandel and Lashof 1974].

2.3. Comparing Two Sets of Measurements—Deming Regression

Ordinary least squares regression cannot be used in the setting where two sets of measurements are compared, because it assumes that only one variable has a measurement error. In order to account for measurement errors in both the independent and dependent variables, we used a functional model and Deming regression, as described by Deal and others [2009]. SAS 9.4 (SAS Institute, Cary, NC) was used for the computation. The Deming regression line of the functional model is estimated by minimizing the sums of squared deviations in both the x and y directions at an angle determined by the ratio of the analytical standard deviations for the two methods. This ratio can be estimated if multiple measurements are made with each method. Alternatively, if only one measurement is made with each method, it can be considered equal to one [Deal and others 2009; Mandel and Lashof 1974]. We used the functional model $E(Yi) = b * E(Xi) + a$, where b denotes the slope and a denotes the intercept and found the relationship and

confidence intervals between the W_{Acti} light and the R, G, and B light responses per light scenario. Moreover, we correlated the average W_{Acti} light C_{VALs} and the average R, G, and B C_{VALs}, respectively, of Halogen versus IllA and CloudySky versus IllA, thus correlating the average C_{VALs} found using a standard calibration method against the average C_{VALs} of the alternative method of calibration.

The software IBM SPSS statistics was used for the computation of statistical estimates of C_{VAL} (average C_{VAL}) using the mixed model procedure.

3. RESULTS

3.1. Lighting Scenarios, Tests, and Effects

The different white light scenarios were characterized by spectrum and illuminance (Fig. 1).

In addition to testing in an illuminant A and illuminant D65 approximate light sources, a fluorescent discharge light source was used with and without a light-diffusing polycarbonate plate in front. The plate resulted in differences in the spatial light distributions between Dif830 and Dir830. The spectra of Dif830 and Dir830 were similar, apart from the small peak around 365 nm appearing in the Dir830 measured spectrum and not in the spectrum of Dif830 (Figs. 1A and 1B).

The distance from each Actiwatch to the light source differed slightly when testing under halogen and fluorescent light sources. Therefore, the light received by the Actiwatches during each calibration test varied (lux variations of approximately 10% for the electrical light sources). Repetitions with random placement of the devices were executed and found to be needed to even out the variations. Thus, every calibration test for one light scenario counted as a repetition and equally in the calculation of the average C_{VALs}.

In darkness, the offsets were found not to be a constant value and therefore an 8-hour period was used to calculate the average. The Actiwatches' average W_{Acti} light offset was between 0.0 and 1.1 (Fig. 2).

As for the W_{Acti} light, some variation was found in the R, G, and B light offsets between the Actiwatches (Figs. 2 and 3). The average R, G, and B light offset of the Actiwatches ranged between 0.000 and 0.184 R light units, 0.000 and 0.088 G light units, and 0.000 and 0.038 B light units (Fig. 3).

Dissimilar C_{VALs} were found depending on the light scenario but also depending on the particular Actiwatch used. The difference in the W_{Acti} light C_{VAL} for the Actiwatches is shown in Figs. 4A–4C.

Whereas the two light intensities of IllA resulted in similar C_{VALs}, the light intensities of IllD65 unfortunately did not (Figs. 4B and 4C). The average W_{Acti} light C_{VALs} found per Actiwatch for Dif830 and Dir830 shows some variation regarding how individual Actiwatches perform relative to the other Actiwatches and depending on the particular light scenario (Fig. 4A). The error bars indicate the 95% confidence intervals with the number of tests included: Dir830 $n = 4$ and Dif830 $n = 8$, where n is the number of calibration tests.

Although the number of tests for Dif830 and Dir830 differs, the sizes of confidence intervals do not differ significantly (Fig. 4A). The number of tests was doubled for the Dif830 (compared with Dir830) to explore why some Actiwatches acted unexpectedly in some Dif830 tests and whether this was just a coincidence resulting in larger variations in C_{VALs} found per Actiwatch. The R, G, and B light C_{VALs} were also found to depend on the particular

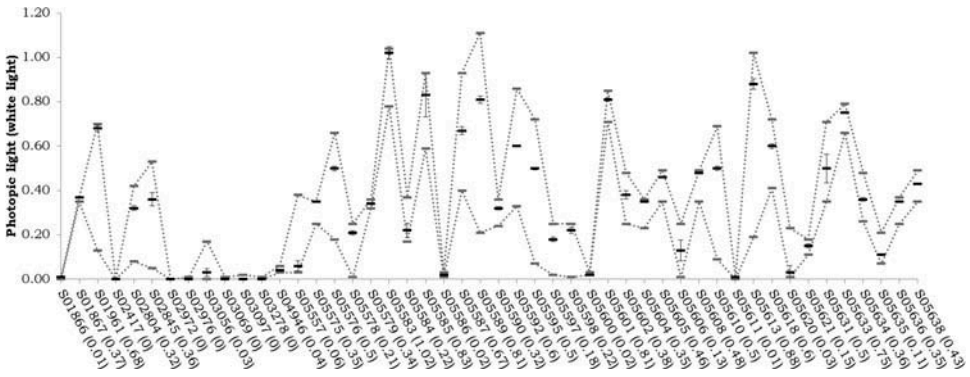


Fig. 2 **White light offset during time of no movement and when Actiwatches are placed in darkness. The white light offset estimate is written in paraphrases after the Actiwatch serial numbers at (the legend) and in the graph indicated with black marks and corresponding standard deviations ($n = 4$; that is, four tests with average white light from 8 hours). The absolute minimum and maximum white light found are indicated by grey marks connected by dashed lines.**

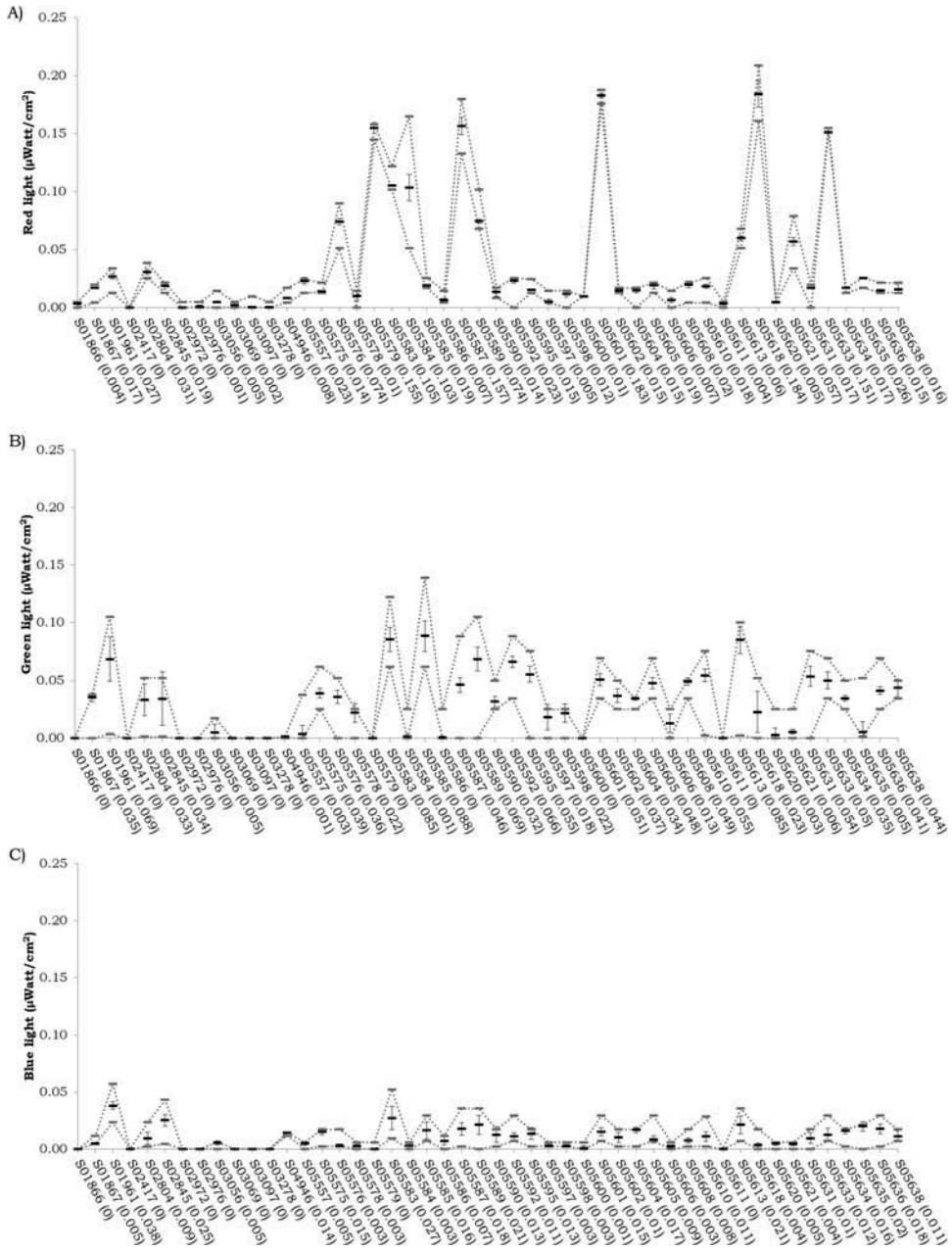


Fig. 3 **Offset of the Actiwatch R, G, and B colors during time when Actiwatches were placed in darkness. The average offset is written in paraphrases after the Actiwatch serial numbers at (the legend) and in the graph indicated with black marks with corresponding standard deviations ($n = 4$; that is, four tests with average light from 8 hours). The absolute minimum and maximum offset found are indicated by grey marks connected by dashed lines.**

Actiwatch used, but some C_{VAL} results of the Dif830 were also found to deviate from the other lighting scenarios (Fig. 5).

Figure 6A shows a plot of W_{Acti} light vs. R, G, and B light of all of the data from the eight Dif830 tests performed, which revealed one reason for the larger variations in C_{VALs} found for Dif830 tests.

When analyzing the data from the Dif830 tests, several significant and sudden drops in the B light (B-light drops) were registered by several Actiwatches (Fig. 5C), although the light present during the tests was stable. These B-light drops affected all of the light channels (the R, G, B, and W_{Acti} light). Data where B-light drops occurred could easily be distinguished from data where these drops did not occur (Fig. 6A). The data with B-light drops caused a 52% reduction of B light and a 2% reduction of the G light, wheras the R light was increased by 23% (Fig. 6A and Table 1). During the eight Dif830 tests, around 18%

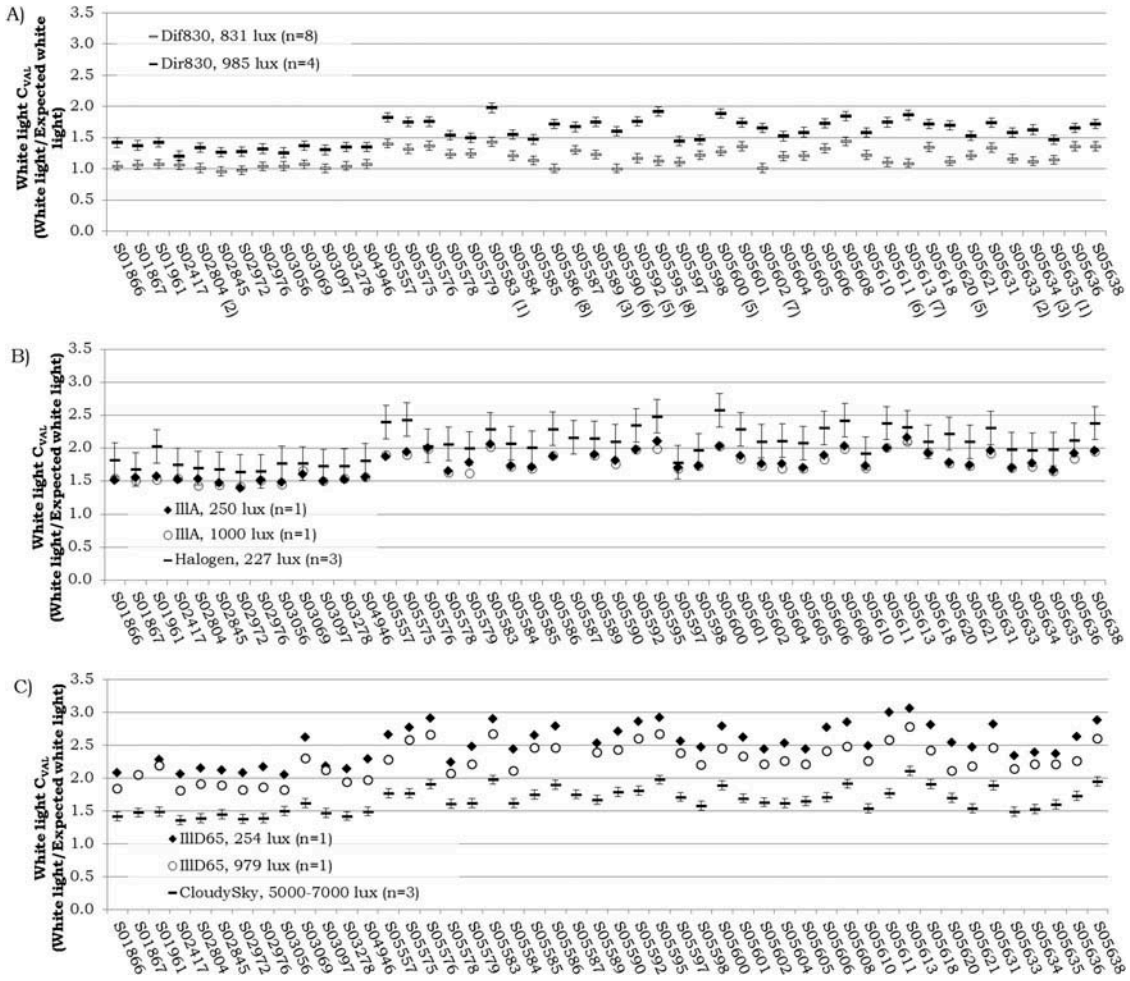


Fig. 4 **Average white light C_{VAL} per Actiwatch for (A) Dif830 ($n = 8$) and Dir830 ($n = 4$); next to the legend in parentheses is written the number of B-light drop incidences during the Dif830 tests, causing the relatively lower C_{VALs}. (B) Halogen ($n = 3$) and IIIA at two intensities. (C) CloudySky ($n = 3$) and IIID65 at two intensities. The error bars indicate the 95% confidence intervals.**

of data showed a trend with B-light drops, where some Actiwatches were more likely to occasionally measure this (see legend in Fig. 4A).

The change in R, G, and B response relative to the RGB sum was found as a reduction of B (−7.33% points) and G (−1.22% points), which equals the increase in R (8.55% points; Table 1). The changed R, G, and B relationship for the data with B-light drops resulted in lower measured W_{Acti} light and thus a reduced average W_{Acti} light C_{VAL} of 1.02 for (selected) B-light drop data versus an average W_{Acti} light C_{VAL} of 1.21 for those categorized as having no B-light drops. The average W_{Acti} light was 854.5 for B-light drop data being significantly lower than the average W_{Acti} light = 1005.3 for the non-B-light drop data (Table 1). For Actiwatches not having any occurrence of B-light drops in any of the eight Dif830 calibration tests, the average W_{Acti} light C_{VAL} was found to be relatively high, whereas those with increasing number of B-light drops had a relatively low average W_{Acti} light C_{VAL} (Fig. 4A).

The correlation between the color responses and W_{Acti} light clearly revealed the B-light drops (Fig. 6A). The B-light drop occurrence could not be ascertained in the other calibration tests (Figs. 6B–6F).

The correlation between G and W_{Acti} light was found to be much stronger and with smaller standard errors compared to those of both R and B when correlated with W_{Acti} light (Fig. 6). The correlations between the Actiwatch colors and W_{Acti} light, where only one light scenario and intensity was used, resulted in intercepts of the regression lines deviating from 0 (Fig. 6). This was more evident under the electrical light sources.

Table 2 lists the overall average values obtained from the different calibrations. The sum of the color responses showed only minor relation to the W_{Acti} light response (Table 2).

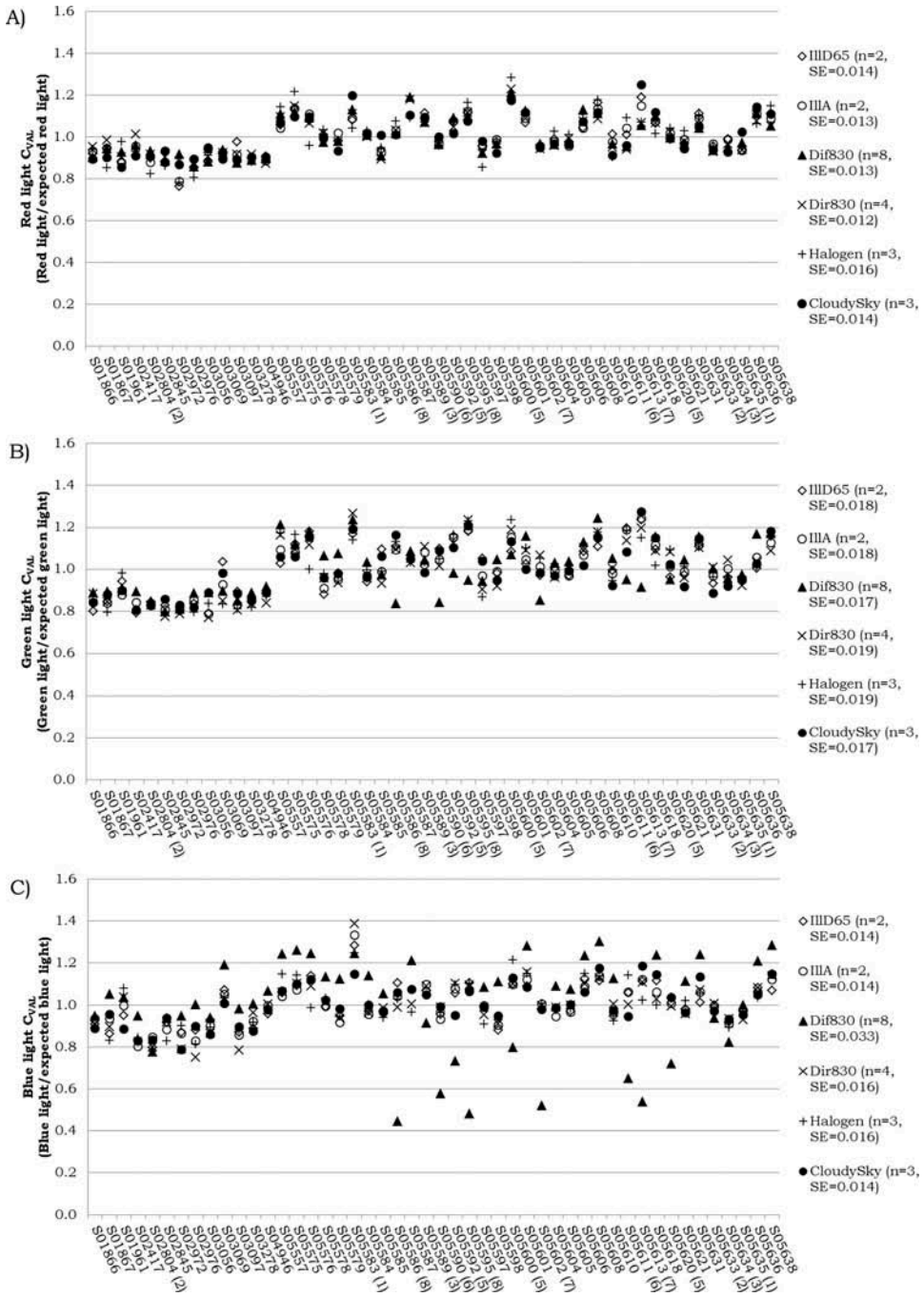


Fig. 5 **The average coloured light response C_{VAL} per Actiwatch for Dif830, Dir830, Halogen, average IIIA of two intensities, CloudySky and average IIID65 of two intensities. (A) R C_{VALs} per Actiwatch. (B) G C_{VALs} per Actiwatch. (C) B C_{VALs} per Actiwatch. The number of calibration tests (n) and SE for the means per light scenario are next to the symbols. Next to the legend in parenthesis is written the amount of B-light drop incidences during the Dif830 tests.**

The G light correlated nicely with the W_{Acti} light, whereas the R and B light did not. Roughly, the G light can be calculated based on measurement of W_{Acti} light and vice versa but only if the light condition is known. The RG and B light versus W_{Acti} light correlation regression slopes differed depending on the light scenario. Therefore, the B and R light responses in particular need individual calibration per Actiwatch. We normalized the R, G, and B light of each individual Actiwatch to the average of all Actiwatches' R, G, and B light response, respectively. This was done for each test and light scenario.

The measurements made using IIIA showed similar C_{VALs} for both of the IIIA intensities (Fig. 4B and Table 2). The levels of C_{VALs} differed for the two IIID65 intensities

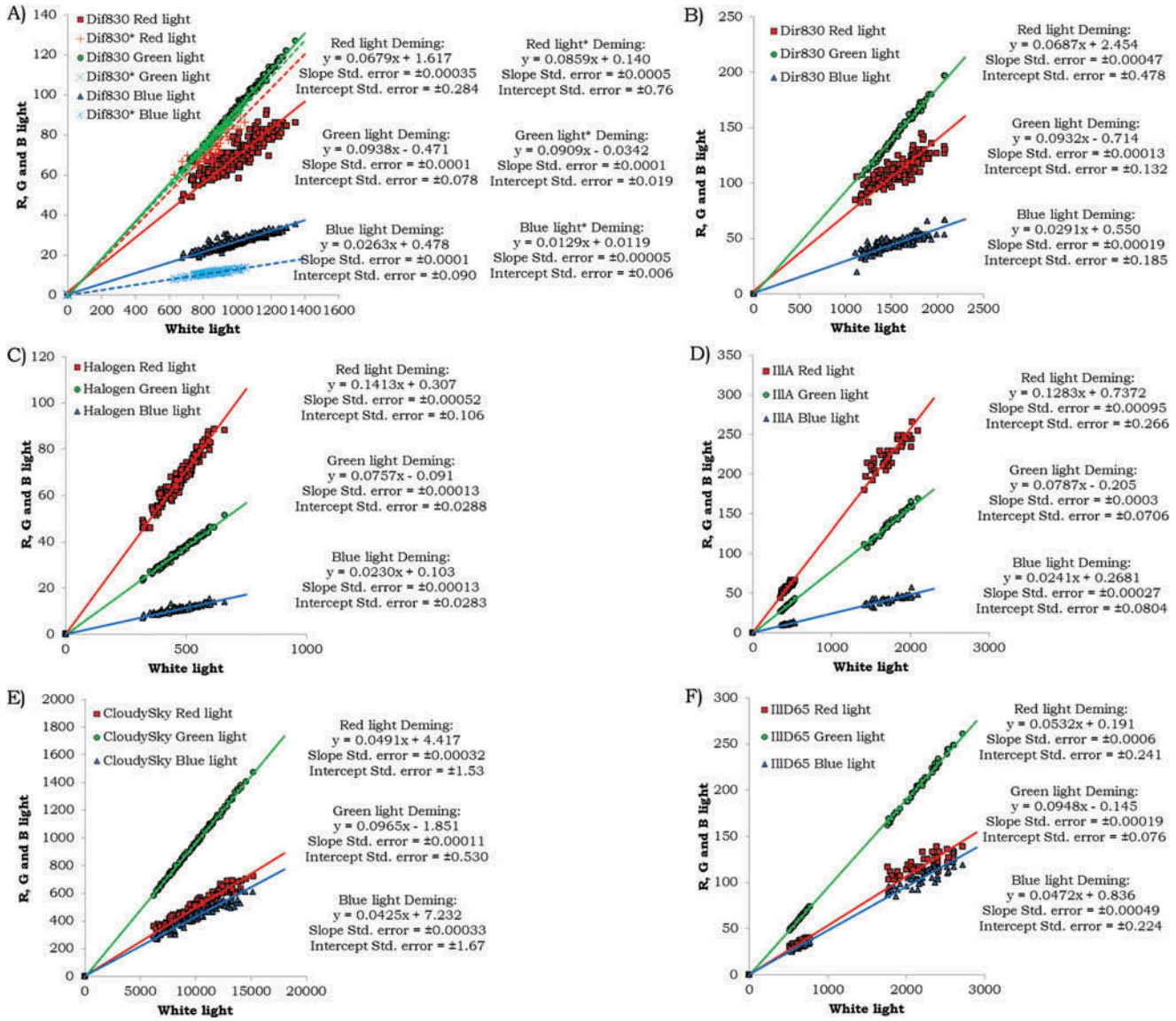


Fig. 6 **Data of white light versus corresponding R, G, and B light from (A) the Dif830, (B) Dir830, (C) Halogen, (D) IIIA, (E) CloudySky, and (F) IIID65 calibration tests. (A) Selected measurements (*) are distinctively different from the rest. During the eight Dif830 tests, 18% of data showed a trend with B-light drops, where some Actiwatches were more likely to measure this occasionally. The slope and intercept of the association and its confidence intervals for the relation between white light and the R, G, and B responses are found using a functional model (Deming regression).**

TABLE 1 The B-light drop occurrence and its influence on the R, G, and B light relationship and the change in the average RGB sum and average white light

	Average R light (% of RGB sum)	Average G light (% of RGB sum)	Average B light (% of RGB sum)	Average RGB sum (%)	Average white light
Dif830 selected (18% B-light drop data)	73.6 (45.38%)	77.6 (47.80%)	11.1 (6.82%)	162.3 (100%)	854.53
Dif830 nonselected (remaining 82% data)	70.2 (36.83%)	93.7 (49.02%)	27.0 (14.16%)	190.9 (100%)	1005.34
Difference (percentage point change)	3.4 (8.55)	−16.1 (−1.22)	−15.9 (−7.33)	−28.6 (0)	−150.81

TABLE 2 Overall average values of reference lux; W_{Acti} light; W_{Acti} light C_{VAL}; RGB sum; and percentage R, G, and B light of the RGB sum, respectively, from the different calibrations. The tests were performed on a total of 48 Actiwatches

Treatment	Repeated tests included	Average reference lux	Average W_{Acti} light	Average W_{Acti} light C_{VAL}	Average RGB sum	RGB sum of W_{Acti} light (%)	R light of RGB sum (%)	G light of RGB sum (%)	B light of RGB sum (%)
Dif830	8	831.0	978.2	1.18	185.8	19.0	38.1	48.9	13.0
Dir830	4	985.2	1562.7	1.59	301.4	19.3	36.7	48.0	15.3
Halogen	3	227.2	468.1	2.06	112.8	24.1	59.0	31.3	9.7
IllA 250 lux	1	249.9	440.8	1.76	102.9	23.3	55.7	33.5	10.8
IllA 1000 lux	1	1000.0	1741.0	1.74	403.8	23.2	55.7	33.8	10.5
CloudySky	3	5885.0	9840.6	1.67	1863.1	18.9	26.2	50.8	22.9
IIlD65 254 lux	1	253.9	639.0	2.52	125.8	19.7	26.7	48.2	25.2
IIlD65 979 lux	1	978.6	2210.0	2.26	433.2	19.6	27.4	48.3	24.3

(Fig. 4C and Table 2). The highest IIlD65 W_{Acti} light C_{VALs} were found when testing at the low IIlD65 intensity and the C_{VALs} were found to be higher than the CloudySky W_{Acti} light C_{VALs}. CloudySky W_{Acti} light C_{VALs} were in closer agreement with the IllA C_{VALs} level (Figs. 4B and 4C and Table 2).

No relationship was found between the battery status of the Actiwatches and the measured light responses and thus the C_{VALs} found (data not shown).

The covering black cloth used to uncover and cover the Actiwatches at every calibration test was found to be necessary. One Actiwatch (S03056) time-keeper was approximately 10 minutes fast after 7 days; that is, after 7 days there were 40 fewer loggings (15-second intervals) stored for S03056 compared to the loggings of the rest of the Actiwatches. Thus, alignment of data was needed. This was found to be important especially for the daylight tests, where the light was not stable (CloudySky).

Daylight calibration at full sunshine was found to result in W_{Acti} light responses exceeding the maximum threshold of the Actiwatches (at 200,000 units W_{Acti} light) and the response being logged was "NaN." One sunlight calibration test where the reference sensor measured 98.5 klux resulted in 12 (out of 48) Actiwatches exceeding the maximum threshold and thus logging NaN (data not shown).

3.2. Comparing Different Calibration Methods

By comparing average C_{VALs} of W_{Acti} light and R, G, and B light, respectively, for Halogen versus IllA and CloudySky versus IIlD65, we found the relationship and confidence intervals between the methods' average C_{VALs} (Fig. 7).

Similarity was found between measurements performed in the two light scenarios, Halogen and IllA, which are basically the same type of light source and show similar spectral compositions (Figs. 1C and 1D). Halogen and IllA showed similar slopes of regression lines and dependence between W_{Acti} light and RGB light responses (Figs. 6C and 6D). The Halogen W_{Acti} light C_{VAL} was found to correlate well with the IllA W_{Acti} light C_{VALs} (Figs. 7A and 7B).

The level of Halogen W_{Acti} light C_{VALs} was somewhat higher than for the IllA. However, relatively large standard errors per Actiwatch were found for Halogen, caused by the varying and relatively short distance to the light source and with imperfect light distribution in the measuring field.

The Actiwatch sensors exhibited a great deal of similarity in measurements made in the two light scenarios CloudySky and IIlD65. This is evident when comparing the slopes of regression lines and the dependence between W_{Acti} light and R, G, and B light plotted for CloudySky and IIlD65, respectively. CloudySky and IIlD65 also showed similar spectral compositions (Figs. 1E and 1F). The CloudySky W_{Acti} light C_{VALs} correlated well with the IIlD65 W_{Acti} light C_{VALs}. Due to the long distance to the light source (the sun) and the high degree of spatially distributed light from the overcast sky of approximately 6 klux at midday, the CloudySky calibration was found to result in small standard errors per Actiwatch.

4. DISCUSSION

4.1. The Need for Device-Specific Correction Factors

A correction factor is usually associated with the reciprocal value of the C_{VAL} used here and thus a multiplication

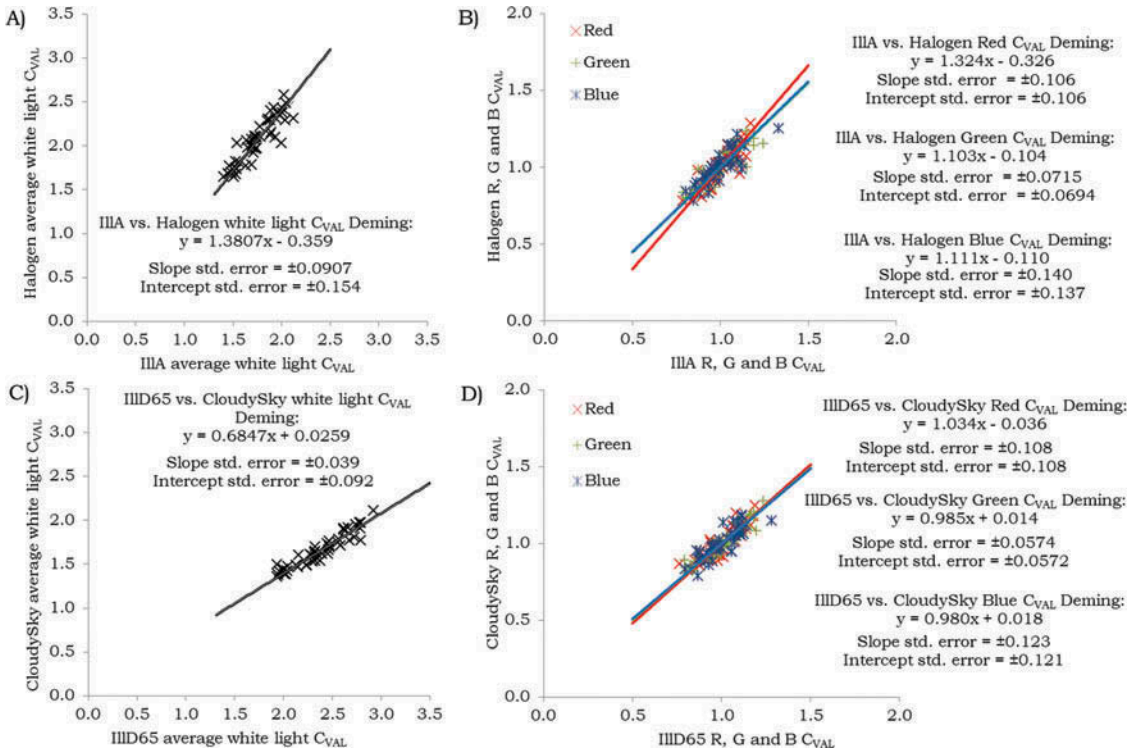


Fig. 7 **Different calibration methods used are compared by plotting the average C_{VAL} found for white, R, G, and B light. (A) Average white light C_{VAL} for IIIA and Halogen; (B) average R, G, and B light C_{VAL} for IIIA and Halogen; (C), average white light C_{VAL} for IIID65 and CloudySky; and (D) average R, G, and B light C_{VAL} for IIID65 and CloudySky. The slope and intercept of the associations and their confidence intervals are found using a functional model and Deming regression [Deal and others 2009].**

value. For simplification of the graphic interpretation, we used least squares regression inclination values (C_{VAL} = measured value/expected value).

It is important that equipment used in research is calibrated and that the light measurements are comparable between devices and across different experiments. The results from testing the Actiwatches clearly demonstrate the need for a field calibration method because large deviations were found between the Actiwatch devices. The need for correction is due to (1) the discrepancy between spectral sensitivity of the sensors and the photopic luminous efficiency function, $V(\lambda)$, and (2) because sensors' spatial responses may not always be cosine corrected. The latter influences the measurements because the light outside laboratory conditions is spatially spread in varying degrees. The variation of the measuring equipment used has to be dealt with. Relying on the calibration made by the manufacturer may not be enough, because equipment variations not dealt with will blur the effects being studied. Sensors used and reporting photopic illuminance are not necessarily cosine corrected and may have a great discrepancy to the $V(\lambda)$. Such sensors will in some cases lead to the wrong conclusions when used in continuously changing spectral and spatial light environments.

4.2. Overestimation of the Amount of Light in the Lighting Scenarios

We performed tests under different white light scenarios including fluorescent light, halogen light, and daylight. The variation in the spectral composition was expected to influence the C_{VAL}s achieved. In a comparative study of three practical field devices, Figueiro and others [2013] reported a discrepancy between $V(\lambda)$ and the spectral response of W_{Acti} light of 83% (f_1' error). They also reported a discrepancy between a cosine distribution and the spatial sensitivity of the Actiwatches' sensors characterized by the f_2 statistics (f_2 error) of 31%, 39%, and 57%, for the R, G, and B sensors, respectively. The consequences are that the photometric measurements of common light sources are biased. When used in the field, Figueiro and others [2013] reported that Actiwatches systematically overestimate the amount of light to which a person is exposed when values are compared with those measured by a Daysimeter device.

In the present study, the Actiwatches' average W_{Acti} light C_{VAL}s was found to differ between the lighting scenarios from 1.18 to 2.52. In the light environments tested and described here, Actiwatches (W_{Acti} light) thus

overestimated photopic illuminance by 18% to 152% depending on the light source and its spectral and spatial light distribution. This is comparable to earlier findings by Figueiro and others [2013]. Figueiro and others [2013] found Actiwatch measurement errors for diffused fluorescent illumination (3500K) and for diffused daylight of 37.1% and 99.9%, respectively.

4.3. Recommendations Concerning Procedures

When devices are tested simultaneously, they cannot be placed at the exact same position. Therefore, the distance to the light source did vary in the side-by-side tests. Light intensity is highly dependent on the distance to the light source, but we handled this by making several repetitions with random placement of the Actiwatches per light scenario and using an average measure. Using light sources for testing that are not stable requires that all measuring devices are tested at the same time. Daylight does change continuously. A relatively long distance to the light source is a major advantage because the relative difference in distances between the simultaneously calibrated devices and the light source becomes smaller. However, the intensity decreases with increased distance. Low intensity is, however, not a problem when using daylight at noon and the distance differences become negligible.

If the measurements are not made under the exact same light scenario, the measurements cannot be compared and used for finding variations between devices. Therefore, ensuring that the logged measurements are synchronized is very important. This was ensured by covering the devices with a black cloth when measurements were not performed and the loggings of light during tests could easily be aligned or if one of the devices had drifted in time as found for one Actiwatch.

That the Actiwatches were lined up and measured perpendicular to the direction to the light source was carefully practiced. The orientation perpendicular to the direction to the sun was of more importance for the calibration results under open sky than the repetitions with randomized placement, giving the directional response of the Actiwatches [Price and others 2012]. Moreover, the cloud cover is important. It was observed that placing the Actiwatches perpendicular to the direction to the sun mattered less during tests with completely overcast sky condition than during tests under only partly overcast sky conditions.

Because some of the Actiwatches (12 out of 48 at 98.5 klux) reached their W_{Acti} light maximum threshold during the sunlight calibration tests, it was not possible to calibrate Actiwatches in sunshine. The CIE standard overcast sky, which is clearly defined and used for calculating daylight factors [CIE 2014; Robbins 1986], is suggested to be the most appropriate to use. However, it is sufficient for the purpose of adjusting inter-equipment variability to test under a diffuse overcast sky.

4.4. Comparing the Calibration Methods, Halogen and IllA

For unknown reasons, the level of Halogen W_{Acti} light C_{VALs} was somewhat higher than for the IllA. This was unexpected considering the spatial characteristics of the light scenarios. The W_{Acti} light C_{VALs} should be highest for the more directed IllA light compared to that of the Halogen light scenario considering the spatial response of the sensors (Figueiro and others 2013; Price and others 2012]. It is unknown whether the divergence is related to the W_{Acti} light C_{VALs} estimations of the IllA or the Halogen calibration and therefore we cannot state that one method is superior to the other. However, correlations were found between the Halogen W_{Acti} light C_{VALs} and the IllA W_{Acti} light C_{VALs}.

4.5. Comparing the Calibration Methods, CloudySky and IllD65

The discrepancy between the level of IllD65 W_{Acti} light C_{VALs} and CloudySky W_{Acti} light C_{VALs} is most likely due to the fact that the light clearly differed between the calibration methods in how spatially distributed the light was. Contrary to the IllD65 calibration, the CloudySky calibration is considered more realistic in the C_{VAL} levels, because the light incidence on the Actiwatches will in practice often be multidirectional. However, there were large differences in illuminance intensities under which the IllD65 and the CloudySky calibration took place. The divergence may be related to both the W_{Acti} light C_{VALs} estimations of the CloudySky and the IllD65 light scenario, and we cannot state that one method is superior to the other. However, correlations were found between the CloudySky W_{Acti} light C_{VALs} and the IllD65 W_{Acti} light C_{VALs}.

4.6. Effects of Light Being Spatially Diffused Using a Frosted Polycarbonate Plate

A frosted polycarbonate plate placed in front of both the Actiwatch and the reference lux sensors in the Dif830 caused a 16% reduction in the average measured reference lux compared with Dir830 (reduction from 985 to 831 lux). In comparison, the Actiwatch sensors measured a reduction in the average W_{Acti} light to be as much as 37% (reduction from 1563 to 978 W_{Acti} light units). This reduction cannot be explained by the B-light drops or the changes in the percentage R, G, and B light of the average RGB sum that only changes slightly. That the relationship between R, G, and B light is more or less unchanged corresponds well with the fact that the measured spectral distribution is reduced equally across the range of visual wavelengths [SABIC Innovative Plastics 2009]. The diffusing effect of the frosted polycarbonate plate and the fact that less light from incident angles not being 90% is received by the Actiwatch sensors [Price and others 2012] compared with the cosine-corrected reference sensor primarily caused the discrepancy in reductions.

Polycarbonate is opaque to all wavelengths below 385 nm [SABIC Innovative Plastics 2009]. The small peak around 365 nm that appears in the Dir830 measured spectrum is not present in the spectrum of Dif830. Combining this with the earlier variability of the RGB sensors' spectral sensitivity of around 20% for the B light at 365 nm [Price and others 2012] may explain the B-light drop phenomenon. In particular, the B light sensor used in Actiwatches seems to be responsive to nonvisual wavelengths (ultraviolet light), and this can explain some of the reduced light measured in the Dif830 treatment by the RGB sensors. For unknown reasons, not all calibration test results for the Dif830 per Actiwatch coincided with each other. If the calibration is performed close to the smallest change in a measured value that the device can detect and the resolution in intensity steps is rough, then this could cause differences in the B light being measured. However, the B light intensity during the Dif830 test was around 10–30 $\mu W/cm^2$, whereas it was around 5–15 $\mu W/cm^2$ during the Halogen tests without any incidence of B-light drops in the Halogen tests. One explanation might be that there are different calibration scales used for the B light signal above and below a certain B light intensity and that the B light intensity split is around 20 $\mu W/cm^2$.

The change in the R, G, and B relationship (B-light drops) cannot explain the major reduction in both the average RGB sum and W_{Acti} light caused by the frosted polycarbonate.

The Actiwatch sensors' directional response is in disagreement with the reference sensors cosine response [Price and others 2012]. The frosted polycarbonate plate that diffuses the light was found to play a major role for the C_{VAL}, and this is the main reason for the reduction from the average Dir830 $C_{VAL} = 1.59$ to the average Dif830 $C_{VAL} = 1.18$, which equals a reduction of 26%.

4.7. Precautions for Straight-line Performance

We correlated the W_{Acti} light and the R, G, and B light, respectively, and we found differing slopes of the Deming regression lines depending on the light environments of the different tests. In the Deming regression analysis we included the offset of the sensors when measuring in darkness as an additional measuring point (Fig. 6). The results show that the measurements deviate to some extent from the regression lines found (Fig. 6). Therefore, the W_{Acti} light most likely depends on the contribution of the devices' R, G, and B wavelength band signals and the light environment. Alternatively, the assumption about the devices having a straight-line performance of the W_{Acti} light to the R, G, and B light may be false. Further tests using different intensities for calibration are needed in order to elucidate whether the devices have a straight-line performance. On the other hand, to use several C_{VALs} depending on the intensity will complicate a field test procedure.

4.8. Differences in the C_{VAL} Found

The two light intensities of IllD65 did not result in the same C_{VALs}, which is most likely related to the reference lux measurement (Fig. 3B). The reference sensor had a relatively large detection area (approximately 7 cm^2) compared to that of the Actiwatch sensor. Adjustment of the light intensity was done by means of a filter and an adjustable collimated lens. During this process, the light distribution on the measuring field of both the Actiwatch sensor and the reference sensor may have changed, causing the difference in C_{VAL} levels of the two IllD65 intensities.

The IllA calibration resulted in W_{Acti} light C_{VALs} being higher than the values found for IllD65. Thus, the spectral power distribution with the low percentage of B light and a high percentage of R light was found to result in lower C_{VALs} and vice versa. However, this is not true in all cases. The relative spectral response of Actiwatch spectrum has earlier been found to be almost insensitive to light emissions between 570 and 600 nm [Figueiro

and others 2013]. For this reason, Actiwatches underestimates photopic illuminance when used under, for example, high-pressure sodium light that still today is used as supplementary plant growth light by many commercial growers. Among other discharge light sources, fluorescent light also has emission lines in this wavelength range. The results shows that though the B light of the RGB sum was higher for Dif830 and Dir830 than for the Halogen and IllA tests, the W_{Acti} light C_{VALs} of Dif830 and Dir830 were lower.

4.9. Spectral Response and Suitable (Night) Applications

Correlations were generally vague for the R and B light, whereas the G light correlated better with the W_{Acti} light. The function of how W_{Acti} light is derived from the R, G, and B responses is unknown and not made public by the manufacturers. Nevertheless, our results suggest that the R and B responses play a minor role compared to the G response, which coincides somewhat with $V(\lambda)$ [CIE 2004] overlapping the spectrum of the G response more than the R and B spectral responses (for figures of the relative spectral sensitivity of Actiwatch R, G, and B sensors, please see Price and others [2012]). However, the G response coincides better with the eye sensitivity function of the scotopic vision regime, $V'(\lambda)$, and peaks around 500 nm, whereas the photopic eye sensitivity function peaks at 555 nm [CIE 2004; Price and others 2012]. The G spectral response of Actiwatches thus coincides better with night vision, although the W_{Acti} light is referred to as photopic.

In addition to light monitoring, personal exposure devices like Actiwatches are able to measure activity and thus sleep quality by registering a person's nightly activity. During nighttime, too much light can reset the circadian clock and therefore light monitoring during the night is often a prerequisite in studies centered on light-initiated circadian biological outcomes. Measuring light at low light intensities is therefore of interest to many circadian scientists studying sleep quality. However, combinations of activity and light measurements in one device can cause some practical difficulties; for example, because blankets (or sleeves) cover the light sensors. For daytime light levels, the W_{Acti} light offsets were found to be negligible with W_{Acti} light responses <1.1 when measuring in darkness. For daytime applications an offset correction will probably not be needed for the tested Actiwatches but it may introduce a measurement bias for low light levels. However, the G sensor spectral sensitivity found by Price and others [2012] is a good prerequisite for light monitoring during the night when light intensities are often low (scotopic vision).

4.10. Manufacturer's Changes of Products

As stated in Actiwatch's specification sheet [Philips Respironics 2013], "Philips Healthcare reserves the right to make changes in specifications and/or to discontinue any product at any time without notice or obligation." Therefore, the data presented here may not be valid, for example, for Actiwatch spectrum devices manufactured at a later date. However, if the measuring product is altered, the need for a simple calibration test when using both new and older types of measuring devices becomes even more relevant for researchers in order to ensure that the devices' measurements are in agreement.

4.11. Differences between the Actiwatches W_{Acti} Light C_{VALs}

Under the same light scenario, the Actiwatches tested differed in measurements of W_{Acti} light. The extreme maximum W_{Acti} light response measured by one Actiwatch was found to be more than 60% higher than the extreme minimum. The calibration tests performed were within illuminance intensities ranging from 250 and 5885 lux. All calibration tests showed the same trend of relatively high and low C_{VAL} values per Actiwatch, despite differences in the spectral composition of light for calibration and the span in illuminances.

As mentioned by Figueiro and others [2013], the manufacturer of Actiwatches states the typical accuracy to be 10% at 1500 lux [Koninklijke Philips 2014]. However, the results presented here clearly show that there is a significant difference of about 60% between Actiwatches tested in our repeated side-by-side tests, which does not correspond to the 10% stated by the manufacturer.

4.12. The R, G, and B Light in Relation to W_{Acti} Light

How the levels of the R and B light were originally set is unknown. Moreover, the units used are not definable but the relationship between RG and B offered by Actiwatches may be used to describe the light in terms of color. This may, if used, reveal new knowledge of the effect of colored and mixed light on well-being issues. However, RGB sensors other than the Actiwatch RGB sensor may most likely result in altered relations of RG and B under the

light scenarios tested. The relations found under selected light scenarios are specific for the Actiwatches.

4.13. Circadian Light Considerations

Light is considered the main human timekeeper and therefore light and health constitute important research fields throughout the world [Aries and others 2013; Czeisler 2013; Eisenstein 2013]. The spectrum of the complex non-image-forming photosensory system involved in the circadian, neuroendocrine, and neurobehavioral responses is not easily definable [Lucas and others 2014]. However, it appears to be most sensitive to blue light of 447–480 nm [Berson and others 2002; Brainard and others 2001; Cajochen and others 2005; Hanifin and Brainard 2007; Iskra-Golec and others 2012; Lucas and others 2014; Thapan and others 2001]. The B light response by Actiwatch spectrum devices peaks in this region [Price and others 2012]. Price and others [2012] suggested that blue light–weighted irradiance data from Actiwatches can be used as a spectral response for circadian studies. According to Price and others [2012], the ratio of 5 B:1 G light Actiwatch readout produces a bell-shaped sensitivity curve that peaks around 480 nm. Nevertheless, photopic illuminance is well defined and is measured in a one-dimensional unit, photopic lux. Hence, photopic illuminance is often used when linking light exposure and health-related issues. We used a photopic illuminance sensor as the reference in our calibration of Actiwatches. However, because of the complexity of the non-image-forming photosensory system and the diversity of broad-spectrum light sensors on the marked, it may still be best to measure the corneal spectral power distribution as suggested by Lucas and others [2014].

5. CONCLUSION

Our results provide a detailed depiction of the 48 Actiwatch devices' responses when calibrated side-by-side. The offset of the W_{Acti} light when measuring in darkness was found to range between 0 and 1.2 lux and thus negligible in most daytime applications but may be accounted for when used in low light intensity studies (Fig. 2).

The variability between the light measurements of the 48 Actiwatches tested was approximately 60% for W_{Acti} light during the calibration tests (Fig. 4). Furthermore, we confirmed earlier findings made by Figueiro and others [2013] that Actiwatches (the W_{Acti} light) often overestimate the actual photopic illuminance (Fig. 4). We found overestimations from 18% to 152% depending on light scenarios tested when compared against a calibrated cosine-corrected photopic illuminance sensor (Table 2).

Being specific for the Actiwatches, we found different relations between R, G, and B light versus W_{Acti} light under selected and described light scenarios. We confirmed earlier findings by Price and others [2012] showing that there is a clear correlation between the W_{Acti} light and G light, whereas the W_{Acti} light correlates to a lesser extent with R light and B light (Fig. 6). The correlation regression slopes found are dependent on light sources and conditions and are specific for the Actiwatch sensors tested.

Differences in the spatial light distribution will affect the correction values (C_{VAL}) when sensors are not cosine corrected and calibrated as described. For Actiwatches, we found that the diffusing effect of a frosted polycarbonate plate in front of the sensors under fluorescent light conditions caused a reduction in C_{VAL} of 26%. This is caused by the fact that only a minimum of light is received at incident angles $> 50°$ by the Actiwatch sensors [Figueiro and others 2013; Price and others 2012].

The CloudySky calibration C_{VALs} correlated with the IllD65 calibration C_{VALs} (Fig. 7C) and the relatively long distance to the light source (the sun) favor use of the CloudySky calibration while resulting in small standard errors (Fig. 4C). In some cases, using more than one light intensity for calibration is needed. Then again, to use several intensities and C_{VALs} per device will complicate a field calibration test procedure.

For comparable analysis and for applications with changing light conditions, we suggest calibrating Actiwatches' and similar light exposure devices' photopic illuminance output under overcast sky conditions against a calibrated cosine-corrected photometer perpendicular to the position of the sun, as described. We consider this field approach to be the best for calibrating light measuring devices of a lower quality.

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